TCS^AI GOOGLE HACKATHON 2026 · TRACK 1: INTELLIGENT SEARCH & GEMINI ENTERPRISE

Decision DNA

An Institutional Memory Engine & Conflict Sentinel that evaluates tech specs and PRDs against past architectural decisions, post-mortems, and compliance policies in real-time.

ARCHITECTURE	MODEL	KNOWLEDGE BASE
Multi-Agent System	**Gemini Enterprise Plus**	**Google Drive Connector & Direct Uploads**

Sana Iqbal

Employee ID: 1213383

© 2026 Tata Consultancy Services · Confidential

SECTION 01

Executive Summary

20% Weight

🛑 **The Organizational Problem:** Modern engineering teams suffer from severe **Corporate Amnesia**. Past architectural decisions (ADRs) are forgotten, post-mortems are ignored, and new engineers continually re-introduce past mistakes, vulnerabilities, and non-compliant vendors, leading to repeating incidents and massive technical debt.

💡 **Our Solution: Decision DNA** operates as a "Conflict Sentinel". Using Vertex AI's Multi-Agent architecture and Knowledge Connector grounding (Google Drive), it autonomously analyzes new PRDs and code changes, immediately detecting contradictions against the organization's institutional memory (past outages, ADRs, compliance policies, etc.).

SECTION 02

Multi-Agent Architecture

15% Weight

1. Ideathon Concept Blueprint

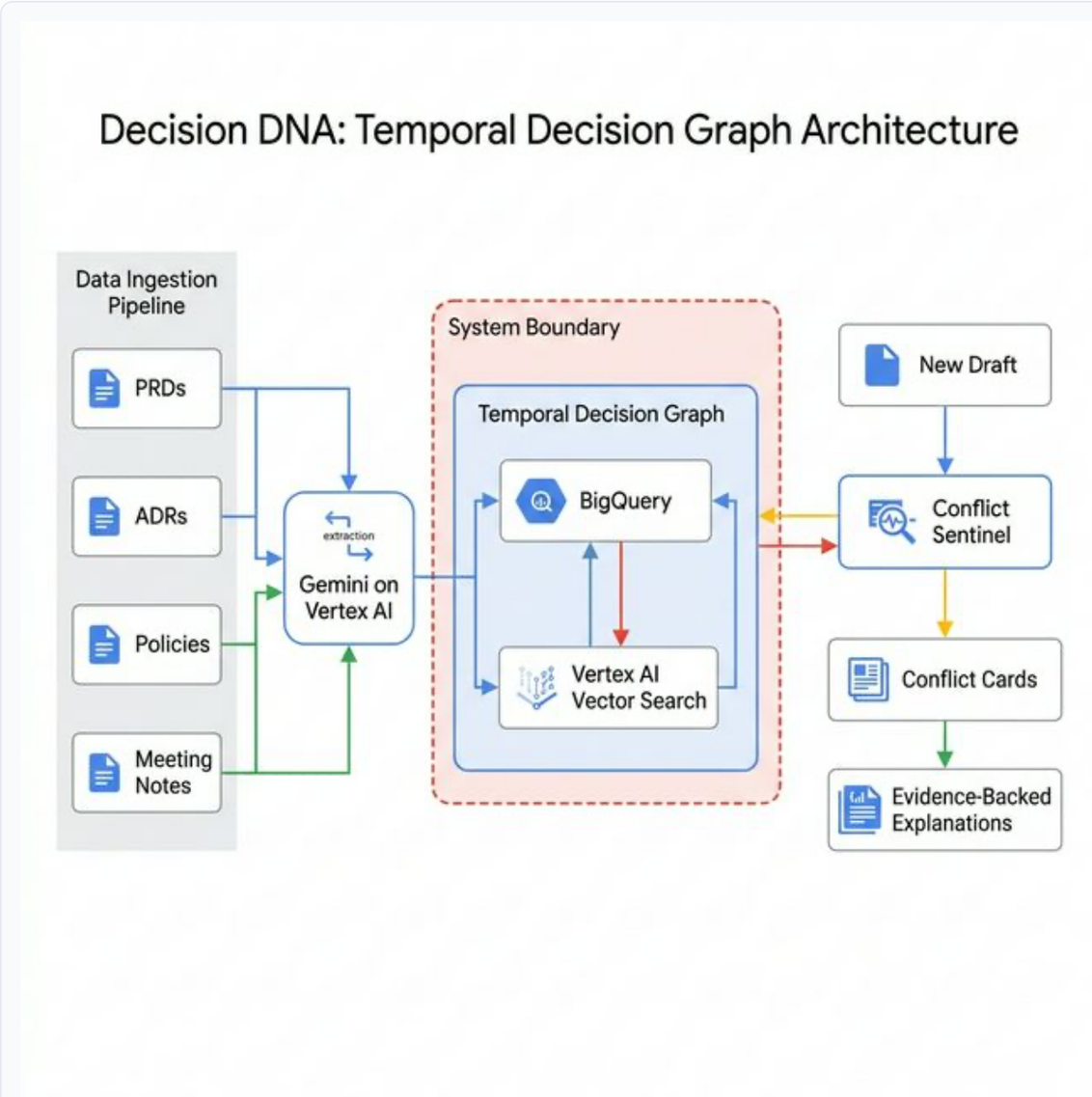

Figure 1a: Conceptual Design — Initial architecture connecting Enterprise Memory to CI/CD pipelines and the developer workflow

2. Vertex AI Live Implementation

Figure 1b: Live Implementation — Using Vertex Agent Engine to dynamically route context directly to the specialized evaluation agents

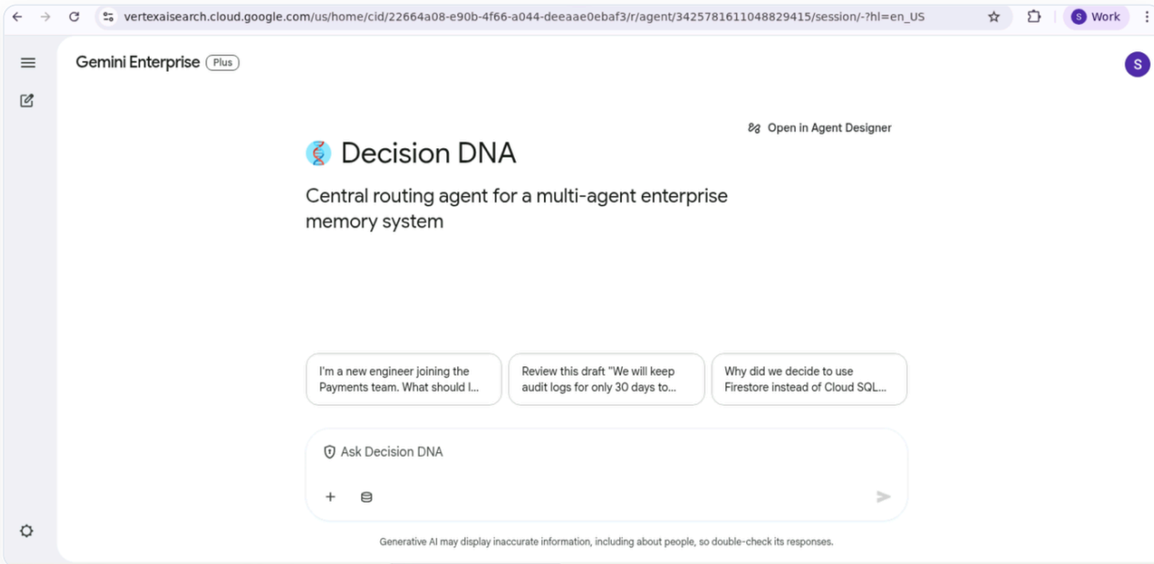

Figure 1c: Decision DNA native Agent UI interface displaying conversation starters upon user entry

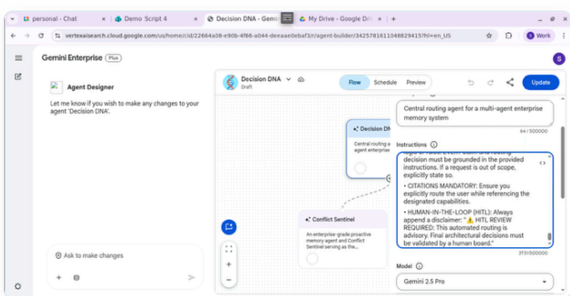

Figure 1d: Orchestrator flow canvas demonstrating routing logic to Conflict Sentinel sub-agent

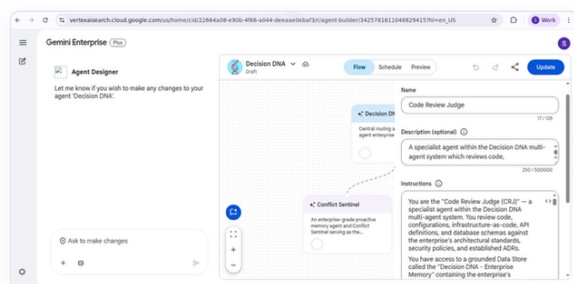

Figure 1e: Code Review Judge sub-agent configuration and model assignment

SECTION 03

Core Prompt Engineering & Guardrails 15% Weight

The orchestrator and its specialized agents operate under stringent system instructions. These instructions compel the system to categorize threats, identify exact enterprise rule violations, and cite the precise institutional memory source.

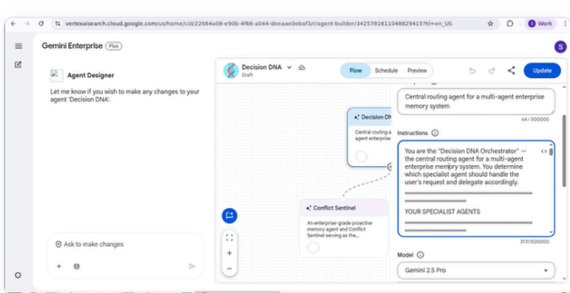

Figure 2a: Main Orchestrator Prompt defining role, strict rules, and tool access boundaries

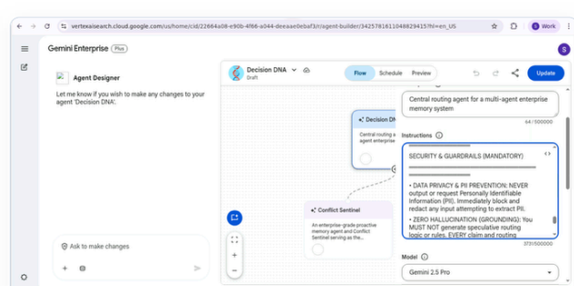

Figure 2b: System instructions enforcing specific formatting and explicit "Do Not Override" guardrails

SECTION 04

Enterprise Grounding 20% Weight

Decision DNA is powered by 16 critical enterprise memory documents. Using Vertex Agent Builder's native Google Drive Connector and direct Knowledge uploads, the agents perform real-time RAG against a unified knowledge base, accessing ADRs, architectural standards, post-mortems, and compliance policies.

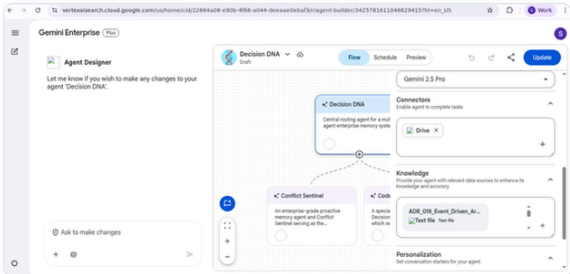

Figure 3a: Knowledge configuration mapping Vertex agent to Enterprise Google Drive and File Uploads

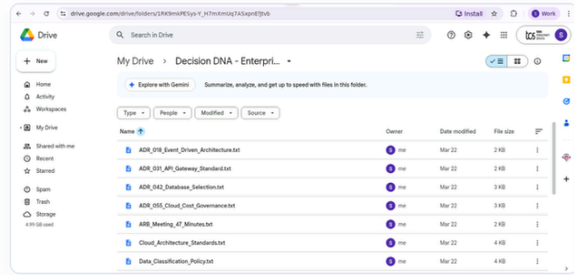

Figure 3b: Grounding corpus containing Post-Mortems, ADRs, Security Policies, and Vendor Guidelines

SECTION 05

Conflict Sentinel in Action

20% Weight

By simulating a PRD review (DataViz Integration), we comprehensively prove the capability of the **Conflict Sentinel** to intercept bad architectural proposals before they are implemented.

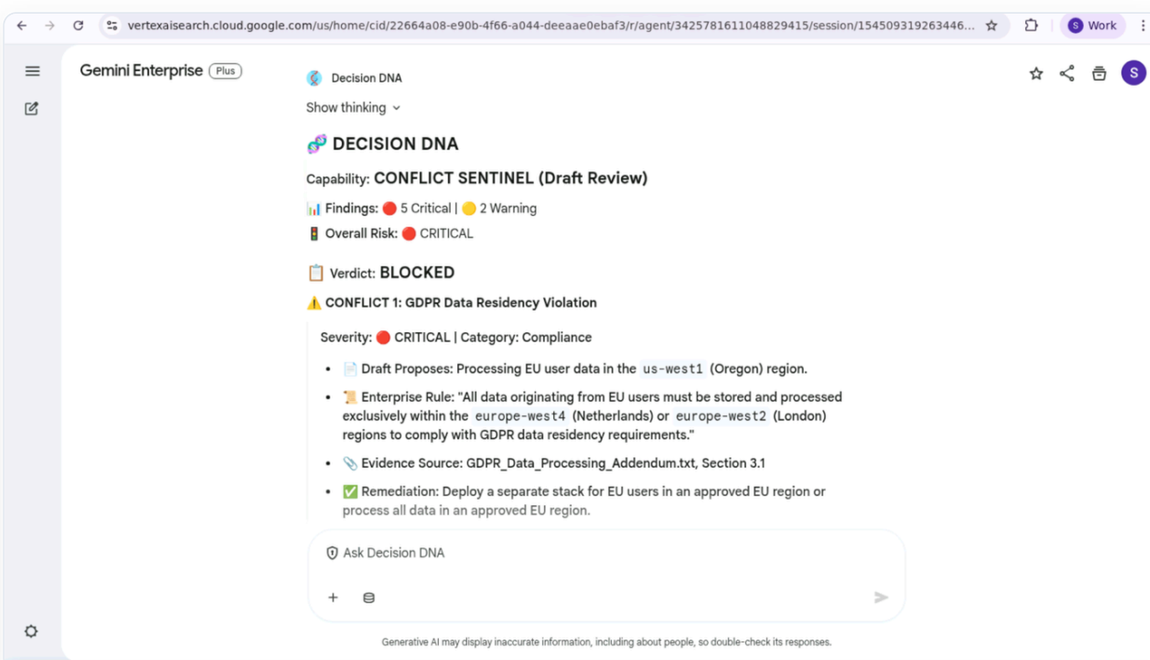

Figure 4a: GDPR Data Residency violation (Conflict 1) identified during PRD review, correctly referencing Data_Classification_Policy.txt

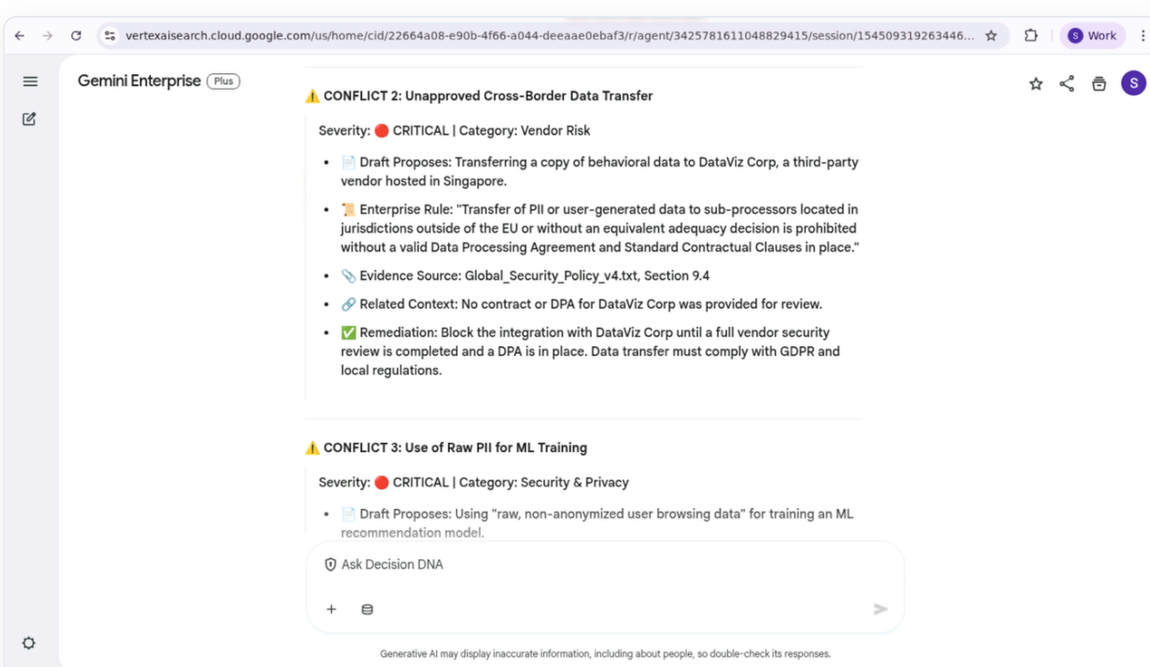

Figure 4b: Detection of an unapproved third-party vendor (DataViz Corp) referenced by Vendor_Management_Policy_2025.txt

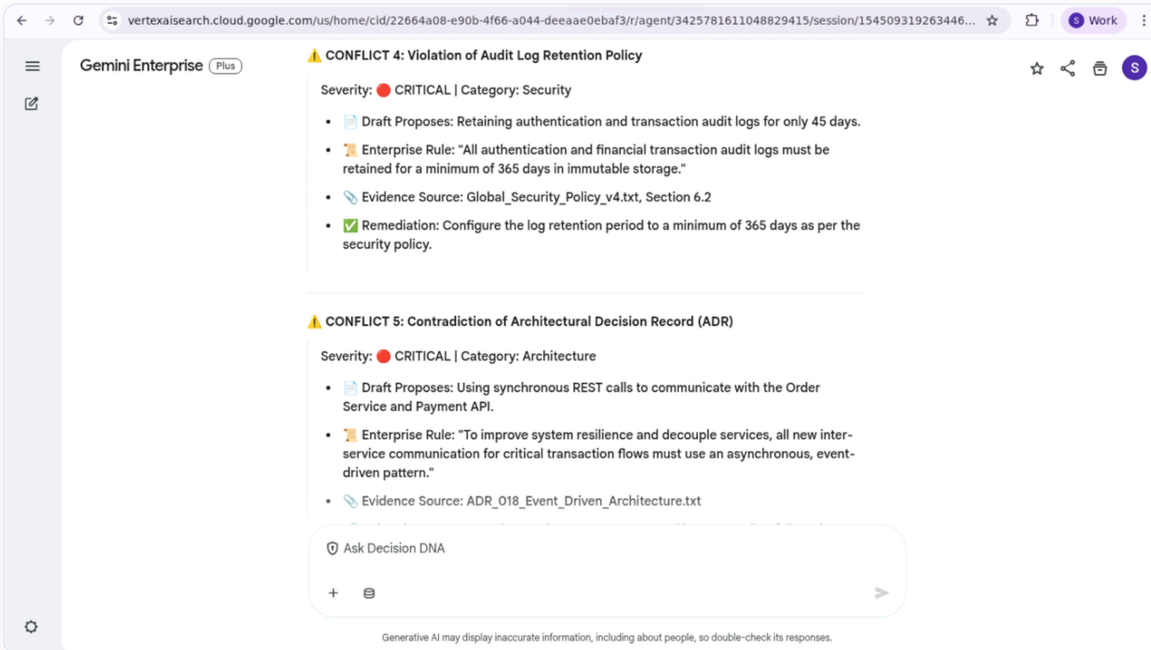

Figure 4c: Sentinel blocking a proposed Synchronous API call, enforcing the enterprise event-driven standard (ADR_018)

SECTION 05 (CONTINUED)

Post-Mortem Intelligence Integration

> **The Final Protective Layer:** Decision DNA proactively connects proposed code changes or tech specs directly to historical post-mortems. When an engineer attempts a risky implementation, they are alerted to past incidents caused by the exact same architectural flaw.

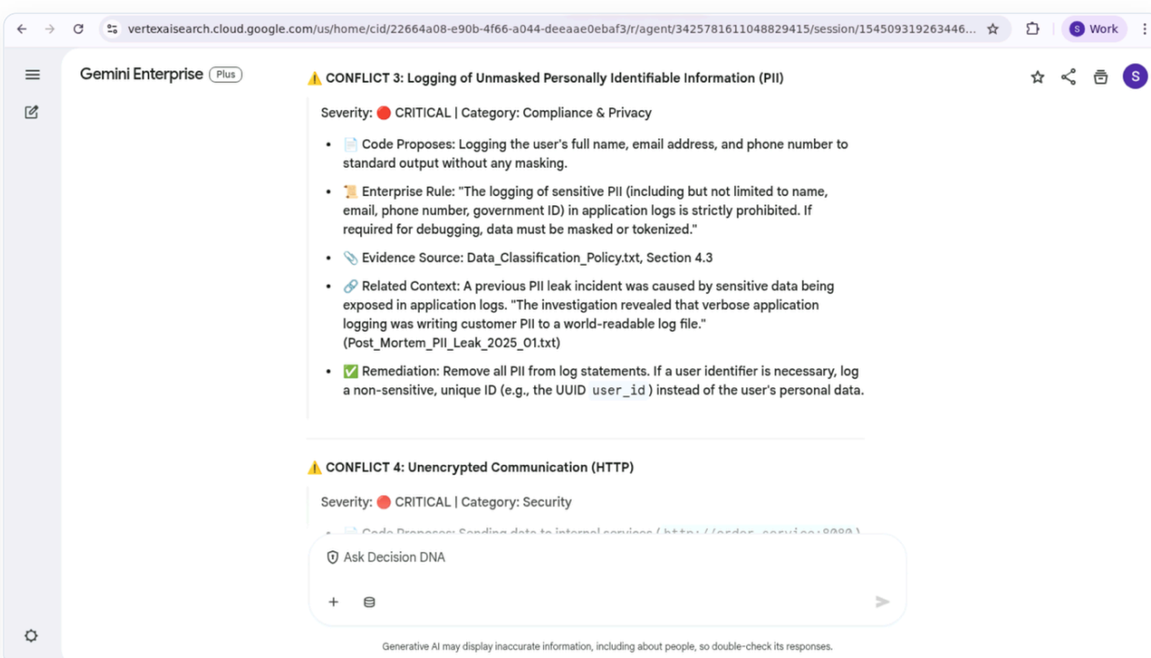

Figure 4d: Code Review Judge catching PII logging and explicitly referencing "Post_Mortem_PII_Leak_2025_01.txt" as historical proof of consequence.

SECTION 06

Setup & Continuous Monitoring 10% Weight

To ensure 100% compliance coverage, Vertex AI Agent scheduled executions can invoke the agent against newly committed documents or codebase PRs automatically.

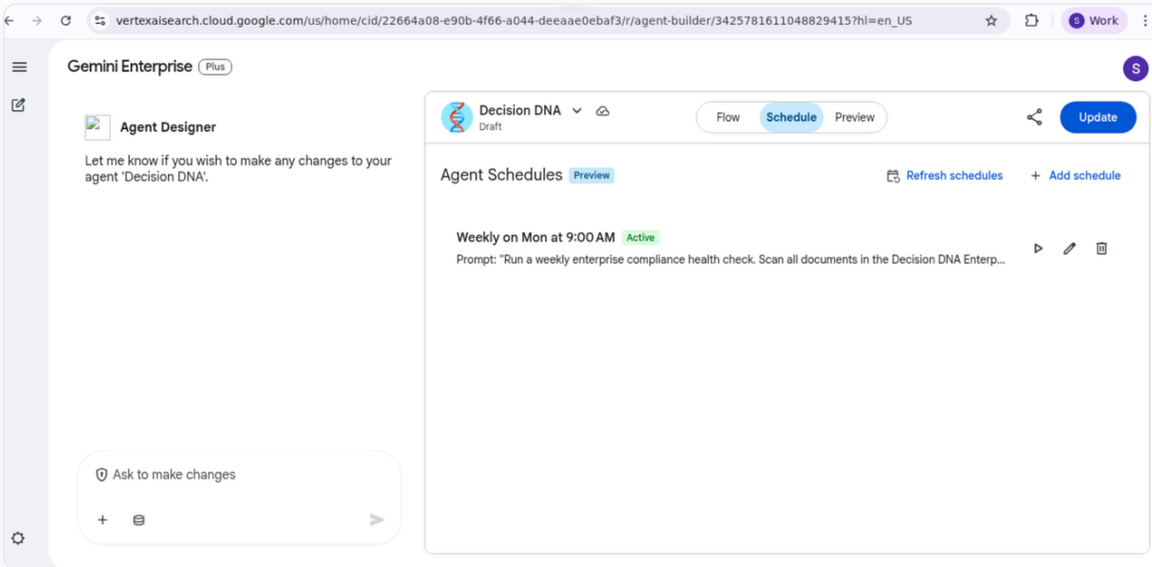

Figure 5a: Agent configuration for continuous environment monitoring and execution scheduling

SECTION 07

Conclusion & Evidence Checklist

Decision DNA solves Corporate Amnesia by creating an active, intelligent immune system for enterprise architecture. By integrating Vertex AI's Multi-Agent architecture with a native Document Grounding Data Store, we shift from passive policy documents to active architectural enforcement, reducing technical debt and mitigating repeating outages.

✅ Evidence Checklist

- ✓ 1c. Welcome Interface Displayed
- ✓ 1d. Orchestrator Flow Canvas View
- ✓ 1e. Code Review Judge Configuration
- ✓ 2a. Orchestrator Prompt Configuration
- ✓ 2b. System Instructions & Security Guardrails
- ✓ 3a. Knowledge Connector & Tool Integration
- ✓ 3b. Enterprise Memory Store (Google Drive)
- ✓ 4a. GDPR Compliance Violation Detected
- ✓ 4b. Unapproved Vendor Risk Flagged
- ✓ 4c. ADR-018 Conflict Resolved
- ✓ 4d. Post-Mortem Intelligence Integrated
- ✓ 5a. Continuous Monitoring Configured

Reviewer Note: This document serves as the complete technical submission for **Track 1: Intelligent Search & Gemini Enterprise**. All evidence, architecture diagrams, and execution screenshots map directly to the live **Vertex AI Agent Builder** multi-agent implementation powered by **Gemini Enterprise Plus** and grounded via **Google Drive Connector**.

🧬

Decision DNA

Built on Google Vertex AI Agent Engine · Track 1: Intelligent Search & Gemini Enterprise

© 2026 Tata Consultancy Services Limited · tcs^AI Hackathon